

ERIC HAY

+1 973-415-9844 ♦ eshay06@gmail.com ♦ www.linkedin.com/in/erichay06

EDUCATION

University of Maryland

Bachelor of Science, Computer Science

Bachelor of Science, Finance

College Park, MD

Anticipated Graduation May 2026

Cumulative GPA: 3.9 / 4.0

Relevant Coursework:

Machine Learning, Artificial Intelligence, Data Science, Data Structures & Algorithms, Web Development, Computer Systems, Object-Oriented Programming, Data Visualization, Linear Algebra, Applied Probability & Statistics

RELEVANT PROJECTS

Data Science, University of Maryland

December 2024

Major Project – Python

- Developed a personalized music recommendation engine using machine learning to analyze the correlation between music genres and mental health indicators.
- Performed extensive exploratory data analysis using pandas and matplotlib to uncover relationships within raw data.
- Designed and built a genre effectiveness scoring algorithm to quantify mental health improvements within listeners.
- Utilized scikit-learn along with custom logic to process data and build a decision tree model which predicts genre of music based on mental health levels.

Organization of Programming Languages, University of Maryland

April 2024

Major Project – OCaml

- Designed and implemented a full lexer, parser, and evaluator system for a custom, well-defined language.
- Transformed raw input strings into a final output through tokenization, syntax analysis, and semantic evaluation.

WORK EXPERIENCE

Barclays, Whippany, NJ

June 2025 – August 2025

Software Engineering Intern

- Built a dynamic UI in React.js with TypeScript to streamline onboarding consumers to enterprise data systems.
- Implemented secure user authentication and authorization by connecting a custom Axios instance to a Spring Boot backend, dynamically rendering UI components based on the user's roles and group permissions
- Connected to AWS DynamoDB supporting CRUD operations for maintaining consumer data.
- Developed a custom view simulation tool to parse raw XML data into a custom table component to facilitate testing.
- Implemented and expanded upon Spring Boot backend functionality to support evolving requirements, including updating DynamoDB keys and exposing new RESTful API endpoints.
- Built and dockerized artifacts using Jenkins and deployed on AWS ECS cluster

Modix 3D, Tel Aviv-Yafo, Israel

June 2024 – July 2024

Software Engineering Intern – Python

- Collaborated with a team to develop innovative software that re-designed the traditional 3D printing process, achieving up to a 35% reduction in print times for large scale prints.
- Conducted extensive research into the hardware and software of 3D printing, including 3D mesh manipulation, G-Code generation, machinery tool path manipulation, and mathematics in order to develop an advanced algorithm.
- Demonstrated strong problem-solving skills and creativity to bring a complex idea to a working prototype solution in a short span of time.
- Documented the software development process and created an in-depth guide for usage of the application.

LEADERSHIP EXPERIENCE

Stifel Investment Banking Case Competition

November 2023

1st Place Winner

- Explored a potential acquisition of Marathon Oil (NYSE: MRO) by Devon Energy Corp (NYSE: DVN).
- Conducted MRO valuation using comparable companies, precedent transactions, and discounted cash flow analysis.
- Compiled our findings into financial models and presented to a potential deal team.

Beta Theta Pi Fraternity, University of Maryland

February 2023 – Present

Executive Member

- Work with a small team to organize all events and meetings within our chapter. Additionally, I am currently the head manager of the chapter house. I work closely with university faculty and our national organization to ensure smooth operation of a house with over 30 residents, including weekly meetings and conferences.
- Develop and harness skills including leadership, communication, team skills, time-management, and planning.

SKILLS

Python (Advanced), React.js (Advanced), C (Advanced), Java (Advanced), TypeScript (Advanced), GCode (Advanced), OCaml (Advanced), JavaScript/HTML/CSS (Advanced), AWS (Intermediate), Node.js (Intermediate), Express.js (Intermediate), Assembly Code (Intermediate), Tableau (Intermediate), D3.js (Beginner)